Routing Evaluation — No Interference

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Contents

1 Evaluation Setup

Identical workloads and scheduler/routing combinations as the `csma_bianchi` interference evaluations, but with `--interference none`: links share bandwidth fairly under concurrent flows but there is no inter-link interference. Results can be compared directly against the interference reports to isolate the cost imposed by wireless contention.

Parameter	Value
Interference	**none** (intra-link fair-share only)
Schedulers	HEFT (calibrated), HEFT-1, HEFT-2
Grid routing	W, S, SH, GS, GC, GB, GO, GSD, GSD-D
Random routing	W, S, SH, GO, GS, GSD, GSD-D
Seeds per combo	30
Grid sizes	4×4 (16 nodes, 40 m spacing) and 7×7 (49 nodes)
Random nodes	50, uniform in $[0, L]^2$, comm range $R = 80$ m
Density levels	L150–L500 ($L = 150$ m to 500 m)
DAGs	Small (8 tasks, fork-join), Large (30/60 tasks, pipeline)

Table 1: Evaluation parameters. All other settings identical to the interference reports.

2 Grid Network Results

Mean makespan in seconds, averaged over 30 seeds. **Bold green**: overall experiment winner. Red: worst in that scheduler column.

2.1 4×4 Grid, Small (8 tasks, fork-join)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	**13.500**	**13.500**	**13.500**
S	**13.500**	**13.500**	**13.500**
SH	**13.500**	**13.500**	**13.500**
GS	**13.500**	**13.500**	**13.500**
GC	**13.500**	**13.500**	**13.500**
GB	**13.500**	**13.500**	**13.500**
GO	**13.500**	**13.500**	**13.500**
GSD	**13.500**	**13.500**	**13.500**
GSD-D	**13.500**	**13.500**	**13.500**
Best	*W 13.500*	*W 13.500*	*W 13.500*

Table 2: 4×4 small (8 tasks, fork-join). Mean makespan over 30 seeds.

2.2 4×4 Grid, Large (30 tasks, 5-stage pipeline)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	130.167	63.667	130.167
S	102.167	63.667	102.167
SH	102.167	63.667	102.167
GS	102.167	63.667	102.167
GC	107.167	63.667	107.167
GB	102.167	68.333	102.167
GO	102.167	63.667	102.167
GSD	70.440	**45.375**	70.440
GSD-D	73.083	**45.375**	73.083
Best	*GSD 70.440*	*GSD 45.375*	*GSD 70.440*

Table 3: 4×4 large (30 tasks, 5-stage pipeline). Mean makespan over 30 seeds.

2.3 7×7 Grid, Small (8 tasks, fork-join)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	**13.500**	**13.500**	**13.500**
S	**13.500**	**13.500**	**13.500**
SH	**13.500**	**13.500**	**13.500**
GS	**13.500**	**13.500**	**13.500**
GC	**13.500**	**13.500**	**13.500**
GB	**13.500**	**13.500**	**13.500**
GO	**13.500**	**13.500**	**13.500**
GSD	**13.500**	**13.500**	**13.500**
GSD-D	**13.500**	**13.500**	**13.500**
Best	*W 13.500*	*W 13.500*	*W 13.500*

Table 4: 7×7 small (8 tasks, fork-join). Mean makespan over 30 seeds.

2.4 7×7 Grid, Large (60 tasks, 6-stage pipeline)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	156.033	141.357	155.471
S	145.100	142.690	145.200
SH	145.333	134.690	145.133
GS	152.071	147.193	153.921
GC	152.908	142.157	152.854
GB	152.358	142.690	154.437
GO	154.058	145.357	153.854
GSD	130.041	**71.483**	128.321
GSD-D	122.935	72.417	136.970
Best	*GSD-D 122.935*	*GSD 71.483*	*GSD 128.321*

Table 5: 7×7 large (60 tasks, 6-stage pipeline). Mean makespan over 30 seeds.

3 Random Network Results

3.1 Topology Statistics

Level	Side L (m)	Links	Avg. degree
L150	150	601	24.0
L200	200	427	17.1
L250	250	309	12.4
L300	300	217	8.7
L350	350	168	6.7
L400	400	146	5.8
L500	500	92	3.7

Table 6: Random network topology statistics (seed 42).

3.2 Makespan vs. Density

Each point is the best-routing mean makespan for that scheduler at that density, averaged over 30 seeds.

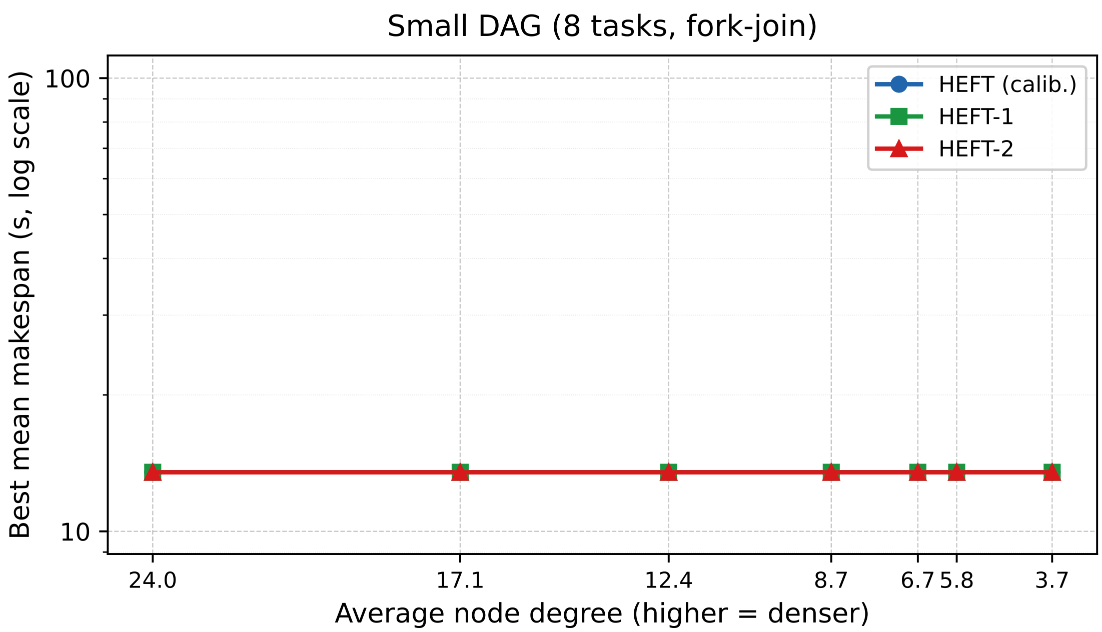

Figure 1: No-interference: best-routing mean makespan vs. average node degree — Small DAG (8 tasks, fork-join).

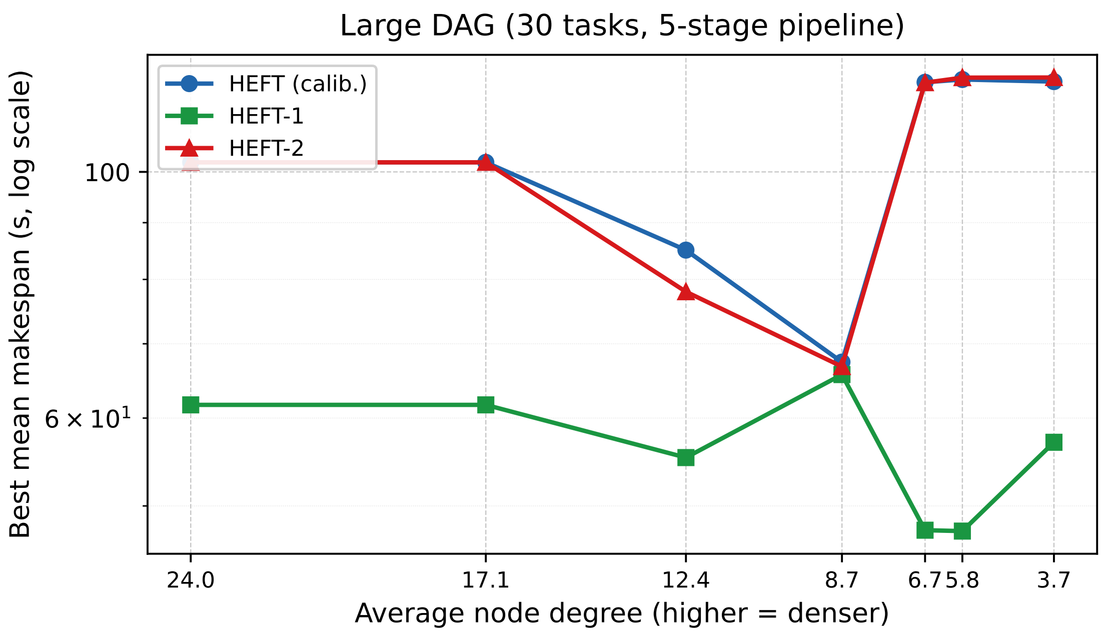

Figure 2: No-interference: best-routing mean makespan vs. average node degree — Large DAG (30 tasks, 5-stage pipeline).

3.3 Best Routing — Small DAG

Density	**Avg. deg.**	**HEFT (calib.) Route (s)**	**HEFT-1 Route (s)**	**HEFT-2 Route (s)**
L150	24.0	W **13.500**	W **13.500**	W **13.500**
L200	17.1	W **13.500**	W **13.500**	W **13.500**
L250	12.4	W **13.500**	W **13.500**	W **13.500**
L300	8.7	W **13.500**	W **13.500**	W **13.500**
L350	6.7	W **13.500**	W **13.500**	W **13.500**
L400	5.8	W **13.500**	W **13.500**	W **13.500**
L500	3.7	W **13.500**	W **13.500**	W **13.500**

Table 7: Best routing scheme and mean makespan per scheduler and density level — Small DAG. **Bold green**: overall best.

3.4 Best Routing — Large DAG (30 tasks)

Density	Avg. deg.	HEFT (calib.) Route (s)	HEFT-1 Route (s)	HEFT-2 Route (s)
L150	24.0	W 102.000	W 61.667	W 102.000
L200	17.1	W 102.000	W 61.667	W 102.000
L250	12.4	GSD-D 85.017	GSD-D 55.278	GSD-D 77.983
L300	8.7	GSD-D 67.394	GSD 65.667	GSD-D 66.767
L350	6.7	S 120.433	GSD 47.544	GSD-D 120.333
L400	5.8	S 121.133	GSD-D **47.458**	W 121.600
L500	3.7	S 120.600	GSD-D 57.061	GSD-D 121.633

Table 8: Best routing scheme and mean makespan per scheduler and density level — Large DAG (30 tasks). **Bold green**: overall best.

4 Full Random Network Results

Mean makespan (seconds) averaged over 30 seeds. **Bold green**: overall best for that density+DAG. Red: worst in that scheduler column.

4.1 L150 (avg. degree 24.0) — Small DAG (8 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	**13.500**	**13.500**	**13.500**
S	**13.500**	**13.500**	**13.500**
SH	**13.500**	**13.500**	**13.500**
GO	**13.500**	**13.500**	**13.500**
GS	**13.500**	**13.500**	**13.500**
GSD	**13.500**	**13.500**	**13.500**
GSD-D	**13.500**	**13.500**	**13.500**

Table 9: L150 ($L = 150\,\mathrm{m}$, avg. degree 24.0) — Small DAG (8 tasks).

4.2 L200 (avg. degree 17.1) — Small DAG (8 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	**13.500**	**13.500**	**13.500**
S	**13.500**	**13.500**	**13.500**
SH	**13.500**	**13.500**	**13.500**
GO	**13.500**	**13.500**	**13.500**
GS	**13.500**	**13.500**	**13.500**
GSD	**13.500**	**13.500**	**13.500**
GSD-D	**13.500**	**13.500**	**13.500**

Table 10: L200 ($L = 200\,\mathrm{m}$, avg. degree 17.1) — Small DAG (8 tasks).

4.3 L250 (avg. degree 12.4) — Small DAG (8 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	**13.500**	**13.500**	**13.500**
S	**13.500**	**13.500**	**13.500**
SH	**13.500**	**13.500**	**13.500**
GO	**13.500**	**13.500**	**13.500**
GS	**13.500**	**13.500**	**13.500**
GSD	**13.500**	**13.500**	**13.500**
GSD-D	**13.500**	**13.500**	**13.500**

Table 11: L250 ($L = 250\,\mathrm{m}$, avg. degree 12.4) — Small DAG (8 tasks).

4.4 L300 (avg. degree 8.7) — Small DAG (8 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	**13.500**	**13.500**	**13.500**
S	**13.500**	**13.500**	**13.500**
SH	**13.500**	**13.500**	**13.500**
GO	**13.500**	**13.500**	**13.500**
GS	**13.500**	**13.500**	**13.500**
GSD	**13.500**	**13.500**	**13.500**
GSD-D	**13.500**	**13.500**	**13.500**

Table 12: L300 ($L = 300\,\mathrm{m}$, avg. degree 8.7) — Small DAG (8 tasks).

4.5 L350 (avg. degree 6.7) — Small DAG (8 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	**13.500**	**13.500**	**13.500**
S	**13.500**	**13.500**	**13.500**
SH	**13.500**	**13.500**	**13.500**
GO	**13.500**	**13.500**	**13.500**
GS	**13.500**	**13.500**	**13.500**
GSD	**13.500**	**13.500**	**13.500**
GSD-D	**13.500**	**13.500**	**13.500**

Table 13: L350 ($L = 350\,\mathrm{m}$, avg. degree 6.7) — Small DAG (8 tasks).

4.6 L400 (avg. degree 5.8) — Small DAG (8 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	**13.500**	**13.500**	**13.500**
S	**13.500**	**13.500**	**13.500**
SH	**13.500**	**13.500**	**13.500**
GO	**13.500**	**13.500**	**13.500**
GS	**13.500**	**13.500**	**13.500**
GSD	**13.500**	**13.500**	**13.500**
GSD-D	**13.500**	**13.500**	**13.500**

Table 14: L400 ($L = 400\,\mathrm{m}$, avg. degree 5.8) — Small DAG (8 tasks).

4.7 L500 (avg. degree 3.7) — Small DAG (8 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	**13.500**	**13.500**	**13.500**
S	**13.500**	**13.500**	**13.500**
SH	**13.500**	**13.500**	**13.500**
GO	**13.500**	**13.500**	**13.500**
GS	**13.500**	**13.500**	**13.500**
GSD	**13.500**	**13.500**	**13.500**
GSD-D	**13.500**	**13.500**	**13.500**

Table 15: L500 ($L = 500\,\mathrm{m}$, avg. degree 3.7) — Small DAG (8 tasks).

4.8 L150 (avg. degree 24.0) — Large DAG (30 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	102.000	**61.667**	102.000
S	102.000	**61.667**	102.000
SH	102.000	**61.667**	102.000
GO	107.000	**61.667**	107.000
GS	107.000	**61.667**	107.000
GSD	107.000	**61.667**	107.000
GSD-D	107.000	**61.667**	107.000

Table 16: L150 ($L = 150\,\mathrm{m}$, avg. degree 24.0) — Large DAG (30 tasks).

4.9 L200 (avg. degree 17.1) — Large DAG (30 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	102.000	**61.667**	102.000
S	102.000	**61.667**	102.000
SH	102.000	**61.667**	102.000
GO	107.000	**61.667**	107.000
GS	107.000	**61.667**	107.000
GSD	107.000	**61.667**	107.000
GSD-D	107.000	**61.667**	107.000

Table 17: L200 ($L = 200\,\text{m}$, avg. degree 17.1) — Large DAG (30 tasks).

4.10 L250 (avg. degree 12.4) — Large DAG (30 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	122.300	61.667	125.100
S	124.267	61.667	122.967
SH	127.467	61.667	128.000
GO	105.000	61.667	103.833
GS	105.000	61.667	103.667
GSD	88.300	55.722	80.083
GSD-D	85.017	**55.278**	77.983

Table 18: L250 ($L = 250\,\text{m}$, avg. degree 12.4) — Large DAG (30 tasks).

4.11 L300 (avg. degree 8.7) — Large DAG (30 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	127.900	71.828	123.067
S	121.867	68.037	122.133
SH	128.500	77.074	124.933
GO	102.833	72.382	103.000
GS	102.500	71.838	102.500
GSD	67.472	**65.667**	68.117
GSD-D	67.394	66.521	66.767

Table 19: L300 ($L = 300\,\text{m}$, avg. degree 8.7) — Large DAG (30 tasks).

4.12 L350 (avg. degree 6.7) — Large DAG (30 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	128.000	61.578	121.200
S	120.433	62.114	122.300
SH	124.167	62.067	126.800
GO	123.567	60.430	125.267
GS	121.200	61.285	125.100
GSD	123.067	**47.544**	121.967
GSD-D	122.467	47.718	120.333

Table 20: L350 ($L = 350\,\text{m}$, avg. degree 6.7) — Large DAG (30 tasks).

4.13 L400 (avg. degree 5.8) — Large DAG (30 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	126.467	60.967	121.600
S	121.133	61.456	122.733
SH	123.400	61.138	124.933
GO	125.867	60.942	125.700
GS	124.333	61.602	126.467
GSD	125.367	47.544	123.067
GSD-D	122.300	**47.458**	125.700

Table 21: L400 ($L = 400\,\text{m}$, avg. degree 5.8) — Large DAG (30 tasks).

4.14 L500 (avg. degree 3.7) — Large DAG (30 tasks)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	122.133	61.194	124.933
S	120.600	60.833	124.600
SH	121.200	60.683	129.000
GO	124.500	61.100	126.033
GS	124.600	61.567	126.733
GSD	123.833	58.050	124.933
GSD-D	125.533	**57.061**	121.633

Table 22: L500 ($L = 500\,\text{m}$, avg. degree 3.7) — Large DAG (30 tasks).

5 Analysis

5.1 Small DAG: Compute-Bound, Routing Irrelevant

The most striking finding is that **all routing schemes and all schedulers produce identical makespan (13.500 s) on the small DAG**, across both grid sizes and all seven random-network density levels. Without interference, the PHY rates on all paths are high enough that transfer time is negligible compared to task compute time (costs 150–1000 cu on nodes at 80–300 cu/s). The 8-task fork-join DAG's makespan is determined entirely by the critical compute path; routing and placement strategy are irrelevant. This is in sharp contrast to the `csma_bianchi` results, where the same DAG showed a 3–12× spread across routing schemes because interference inflated transfer times to the point where they dominated the makespan.

5.2 Large DAG: HEFT-1 Advantage Shrinks, GSD Emerges

On the large DAG the picture is more nuanced:

- **HEFT-1 still wins, but by a much smaller margin.** On the 4×4 grid, HEFT-1/GSD achieves 45.4 s vs. HEFT/GSD = HEFT-2/GSD = 70.4 s—a 1.6× advantage. Under csma_bianchi the same experiments showed 3–7×. The reduction confirms that co-location's main benefit was avoiding inter-link spectrum contention, not intra-link queuing.
- **GSD is the best routing scheme on grids.** GSD (interference-aware dynamic) wins in all four grid experiments under HEFT-1, including the 7×7 large (71.5 s), where under interference GO had won (542 s). Without interference GSD's dynamic path selection still reduces intra-link congestion by spreading flows across the grid. GSD-D wins only for the 7×7 large under HEFT (122.9 s).
- **Widest-path (W) is no longer the worst.** Under interference W was always worst by a large margin (e.g. 5696 s vs. 542 s for 7×7 large). Without interference W is within 2× of GSD on grids (141 s vs. 71 s under HEFT-1 on 7×7 large), confirming that W's penalty came almost entirely from inter-link contention.

5.3 Random Networks: Flat Small DAG, Density-Insensitive Large DAG

On random networks without interference:

- **Small DAG:** All (scheduler, routing, density) combinations produce 13.500 s—exactly as on the grids. Compute dominates.
- **Large DAG:** HEFT-1 achieves 47–66 s across all density levels with the best routing scheme (GSD or GSD-D), compared to 150–927 s under csma_bianchi. The makespan is *nearly flat* across densities: without interference, denser topologies provide more path diversity for GSD's dynamic selection, but offer no dramatic improvement because compute time already dominates.
- **GSD and GSD-D win at all densities** under HEFT-1 for the large DAG (GSD-D at L250–L400 and L500; GSD at L350). This contrasts with the csma_bianchi case where SH dominated at high density and W/GSD-D won only at very sparse topologies.
- **SH never wins.** Without the interference-reduction rationale, SH offers no advantage over S or W. GSD/GSD-D consistently outperform SH by 10–30% on the large DAG.
- **HEFT and HEFT-2 tie at most densities** (W or S gives the same result for both), matching the pattern seen on the grid where calibration only diverges when topology is large.

5.4 Key Takeaways

1. **Interference is the dominant cost on these workloads.** The small DAG shifts from a 13× spread across routing schemes (with interference) to a flat 13.5 s (without). Interference, not bandwidth or hop count, was the binding constraint.

2. **HEFT-1 co-location benefit is mostly interference avoidance.** The 3–7× advantage over HEFT/HEFT-2 shrinks to 1.6× without interference. On interference-free networks, spreading tasks is not nearly as costly.

3. **GSD replaces SH as best routing for large DAGs.** Without interference, GSD's dynamic load spreading beats SH's hop-count minimisation because relay hops no longer carry an interference penalty.

4. **W is rehabilitated.** Widest-path routing was catastrophic under csma_bianchi because it routed flows through the busiest, most interfered links. Without interference it is within 2× of GSD, and is the best scheme for several (density, scheduler) combinations on the large random-network DAG.

5.5 Reproducing These Results

```
cd ncsim/
python run_no_interference_eval.py
# JSON: /tmp/ncsim_no_interference_eval/no_interference_results.json
```